

EV Battery Management System

Dashboard Study Guide

A complete guide to understanding every part of the BMS dashboard

July 2026

Chapter 1: What is a Battery Management System (BMS)?

A Battery Management System (BMS) is an electronic system that manages and monitors a rechargeable battery pack in an Electric Vehicle. It is the brain of the EV battery — without it, the battery could overheat, overcharge, or get damaged.

1.1 Why is BMS Important?

- Prevents overcharging and over-discharging of the battery
- Monitors battery temperature to prevent thermal runaway (fire risk)
- Tracks the State of Charge (SOC) — how much energy is left
- Tracks the State of Health (SOH) — how degraded the battery is
- Balances individual cells so all cells age equally
- Provides data for analytics and predictive maintenance

1.2 Key BMS Parameters

SOC (State of Charge): Think of it like a fuel gauge. 100% means fully charged, 0% means empty. In our dashboard, SOC is shown as a percentage and plotted over time to see how it changes during driving, charging, and parking.

SOH (State of Health): Measures how much capacity the battery has compared to when it was new. A brand new battery has 100% SOH. Over years of use, it degrades. Our vehicle EV001 shows ~96.5% SOH, meaning the battery has lost about 3.5% of its original capacity.

Voltage: The electrical pressure of the battery pack, measured in Volts (V). Higher SOC means higher voltage. In our data, voltage ranges from ~330V to ~400V.

Current: The flow of electricity, measured in Amperes (A). Positive current means the battery is discharging (driving). Negative current means charging.

Temperature: Battery temperature in degrees Celsius (°C). Batteries perform best between 20-40°C. Above 55°C is dangerous.

Power: Calculated as Voltage × Current, measured in kilowatts (kW). It tells you how much energy is being consumed or supplied per second.

Speed: Vehicle speed in km/h. It is zero during parking and charging.

Chapter 2: The Three Operating Modes

An EV operates in three distinct modes throughout the day. Each mode has very different battery behaviour.

2.1 Driving Mode

- The vehicle is moving — Speed > 0
- Current is positive — battery is discharging (giving energy to the motor)
- SOC decreases gradually as energy is consumed
- Voltage drops slowly as SOC decreases
- Temperature increases due to the electrical load on the battery
- Power is positive and varies with driving intensity
- Charging Status = No

In our dataset, the vehicle drives during morning commute (06:30), errands, lunch trip, afternoon drive, evening commute, and a short night trip. Total driving time is approximately 322 minutes.

2.2 Charging Mode

- The vehicle is stationary — Speed = 0
- Current is negative — electricity is flowing INTO the battery
- SOC increases as the battery fills up
- Voltage increases slightly as SOC rises
- Temperature increases slightly due to internal resistance heating
- Charging Status = Yes

Our dataset has two charging sessions: one in the afternoon (15:07-16:22) and one in the evening (18:37-19:37). Total charging time is approximately 135 minutes.

2.3 Parking Mode

- The vehicle is stationary — Speed = 0
- Current is approximately zero — no significant energy flow
- SOC stays almost constant (tiny self-discharge)
- Voltage remains stable
- Temperature slowly drifts toward ambient temperature (~28°C)
- Charging Status = No

Parking is the dominant mode — about 983 minutes of the day. The battery is essentially resting during this time.

Chapter 3: Understanding the Dashboard

3.1 Sidebar

The sidebar on the left contains the mode selector dropdown with four options: All, Driving, Charging, and Parking. Below the dropdown, it shows the Vehicle ID, the date, and total record count. When you change the mode, all KPIs and charts update instantly.

3.2 KPI Cards

Five metric cards appear at the top of the dashboard. They show the average values for the currently selected mode:

- Avg SOC (%) — average battery charge level
- Avg SOH (%) — average battery health
- Avg Voltage (V) — average pack voltage
- Avg Temp (°C) — average battery temperature
- Avg Power (kW) — average power consumption/supply

When you switch to 'Driving' mode, these KPIs recalculate using only driving data. This lets you compare how different modes affect battery performance.

3.3 Charts — All Mode

When 'All' is selected, four charts appear:

- SOC vs Time — shows the full-day charge/discharge pattern. You can see SOC dropping during driving, rising during charging, and staying flat during parking.
- Temperature vs Time — shows how battery temperature changes. It rises during driving and charging, then cools during parking.
- Voltage vs Time — mirrors the SOC pattern since voltage is directly related to charge level.
- Mode Distribution (Pie Chart) — shows the proportion of time spent in each mode. Parking dominates since the vehicle is parked overnight.

3.4 Charts — Driving Mode

When 'Driving' is selected, four driving-specific charts appear:

- SOC vs Time — shows how charge depletes during each driving segment
- Speed vs Time — shows vehicle speed variation (15-80 km/h)
- Power vs Time — shows energy consumption intensity
- Temperature vs Time — shows heat buildup during driving

3.5 Charts — Charging Mode

When 'Charging' is selected:

- SOC vs Time — shows the battery filling up
- Voltage vs Time — shows voltage rising as SOC increases
- Charging Current (A) — shows the absolute current flowing into the battery
- Temperature vs Time — shows slight temperature rise during charging

3.6 Charts — Parking Mode

When 'Parking' is selected:

- Voltage Stability — should be a nearly flat line, confirming no drain
- Temperature Trend — shows gradual cooling toward ambient temperature
- SOC Stability — should be nearly flat, confirming minimal self-discharge

3.7 PDF Report

At the bottom of the dashboard, a summary report table is displayed. You can download a detailed PDF report by clicking the download button. The PDF contains:

- Executive summary with key statistics
- A metrics table with all averages, min/max values, and record counts
- Mode-wise breakdown table
- Detailed analysis for each mode (Driving, Charging, Parking)
- Battery health summary

Chapter 4: How to Interpret the Data

4.1 Reading the SOC Curve

The SOC curve is the most important chart in BMS analytics. Here is how to read it:

- Downward slopes = driving (battery discharging)
- Upward slopes = charging (battery filling)
- Flat segments = parking (battery resting)
- Steeper slope = more aggressive driving or faster charging
- The SOC should never drop below 10% (deep discharge damages the battery)
- The SOC should never stay at 100% for too long (reduces battery life)

4.2 Temperature Monitoring

Battery temperature is critical for safety:

- Below 0°C — battery performance drops, charging should be slow
- 20-40°C — optimal operating range
- 40-50°C — elevated but acceptable during heavy use
- Above 55°C — DANGER zone, risk of thermal runaway

In our data, the temperature stays between 28-34°C, which is perfectly safe for operation in Pune's climate.

4.3 Current Sign Convention

Understanding the current sign is essential:

- Positive current (+) = battery is DISCHARGING (energy flows out to the motor)
- Negative current (-) = battery is CHARGING (energy flows into the battery)
- Near-zero current = PARKING (no significant energy flow)

4.4 Power Calculation

Power = $|\text{Voltage} \times \text{Current}| / 1000$ (in kilowatts). During driving, power tells you how hard the motor is working. During charging, it tells you how fast the battery is being replenished. During parking, power should be nearly zero.

Chapter 5: Key Formulas and Concepts

- 1. Power (kW):** $P = |V \times I| / 1000$, where V = Voltage (V), I = Current (A)
- 2. Energy consumed:** $E = P \times t$, where P = Power (kW), t = time (hours)
- 3. SOC change:** If SOC drops from 85% to 73%, the battery used 12% of its total capacity
- 4. Range estimation:** If 12% SOC covers 50 km, then 100% SOC covers approximately $50/12 \times 100 = 416$ km
- 5. Charging rate:** If SOC rises from 73% to 85% in 135 minutes, charging rate = $12\%/2.25h = 5.3\%/hour$

Chapter 6: Summary

This dashboard gives you a complete picture of how an EV battery behaves throughout a day. By studying the SOC curve, temperature patterns, current flow, and power consumption across different modes, you can understand battery performance, identify potential issues, and make informed decisions about charging strategy and driving habits.

The PDF report provides a permanent record of the analysis that can be shared with professors, colleagues, or included in project documentation.